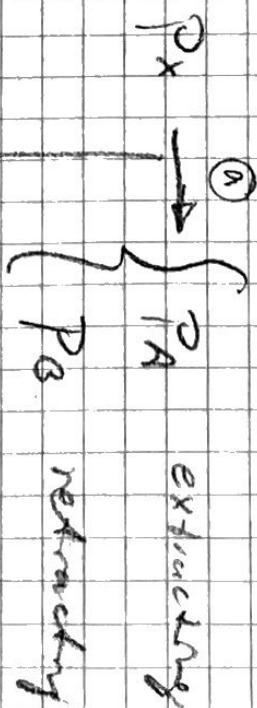


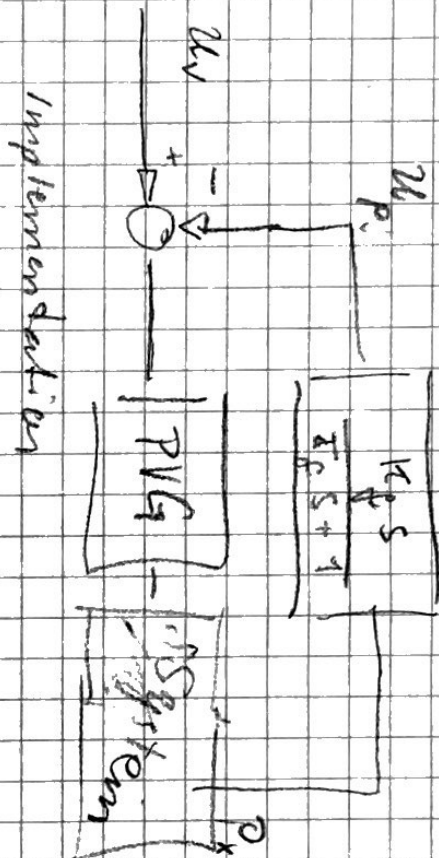
T_f

$$T_f \in [0.05 \dots 0.17 \text{ s}]$$

K_f ? T_{ry} , but small



area = $\frac{\pi \cdot r^2}{2} < 1$



Implementation

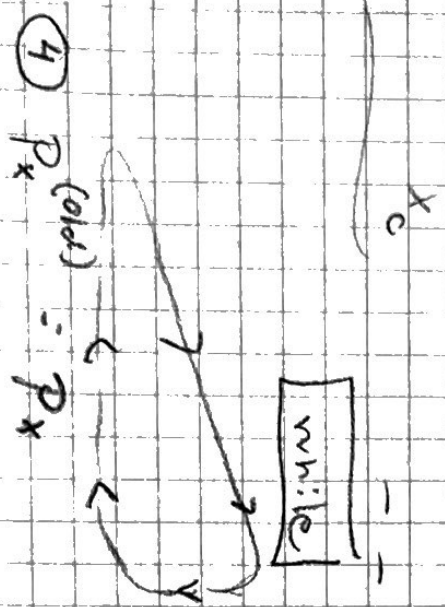
$$u_p = 0 \quad \leftarrow \text{Initialization}$$

$$p_x^{(old)} = p_x^{(ss)}$$

$$\textcircled{1} \quad \dot{p}_x = (p_x - p_x^{(old)}) / \Delta t$$

$$\textcircled{2} \quad \dot{u}_p = K_f \cdot (\dot{p}_x - \dot{u}_p) / T_f$$

$$\textcircled{3} \quad u_p = u_p + \dot{u}_p \cdot \Delta t$$

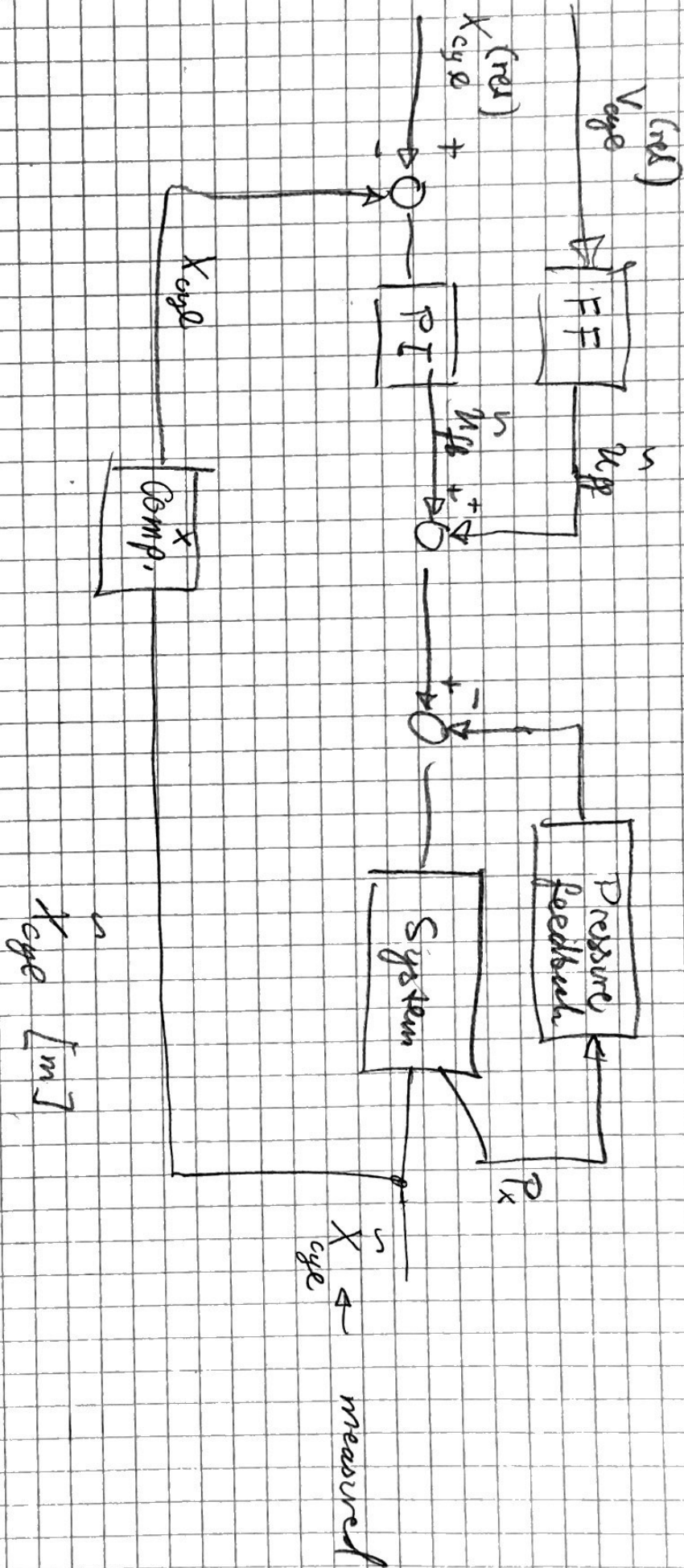


FEED FORWARD

FEED PI BACK

4

Feed back loop

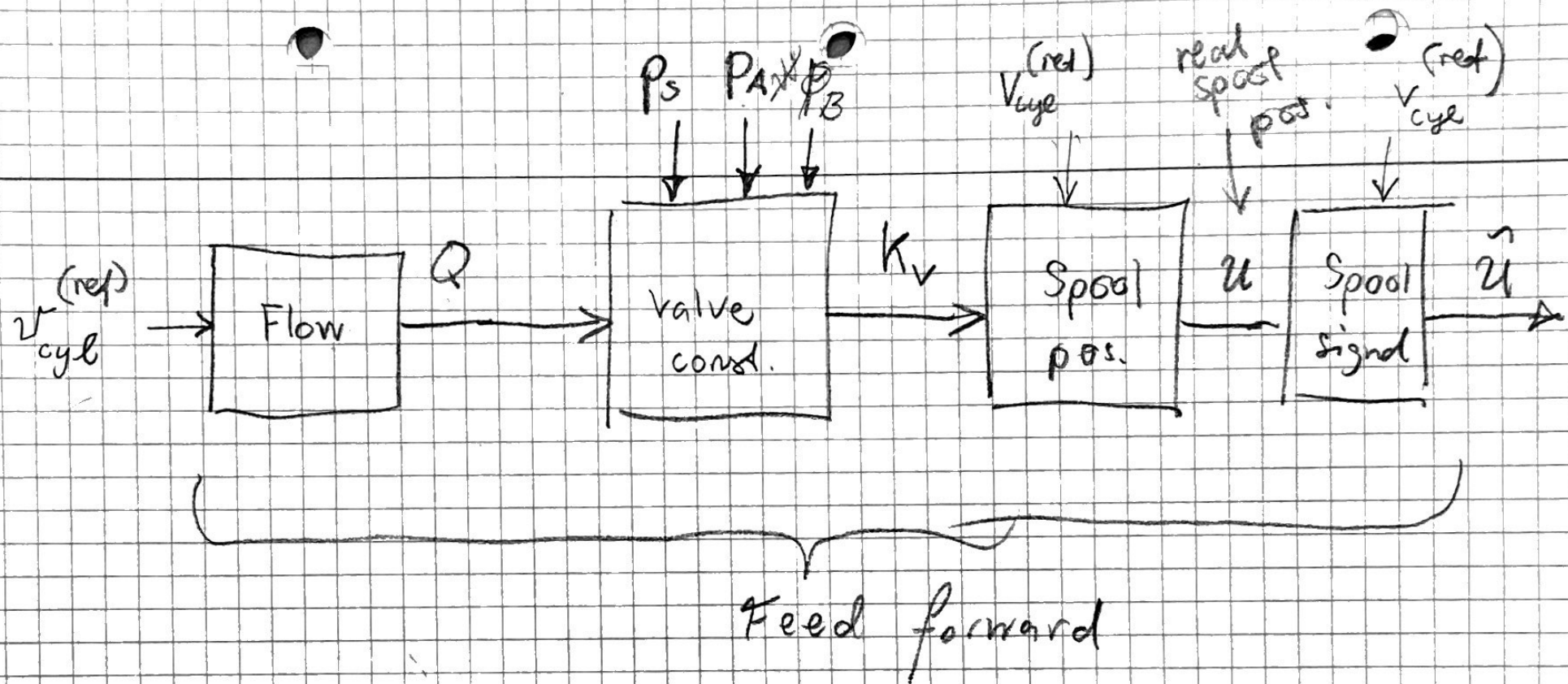


X_{Comp}

$$X_{cye} = X_{cye}$$

$$-0.0006 - 0.0104 \cdot X_{cye} + 0.5855 \cdot X_{cye}^2 - 1.1291 \cdot X_{cye}^3$$

(1)



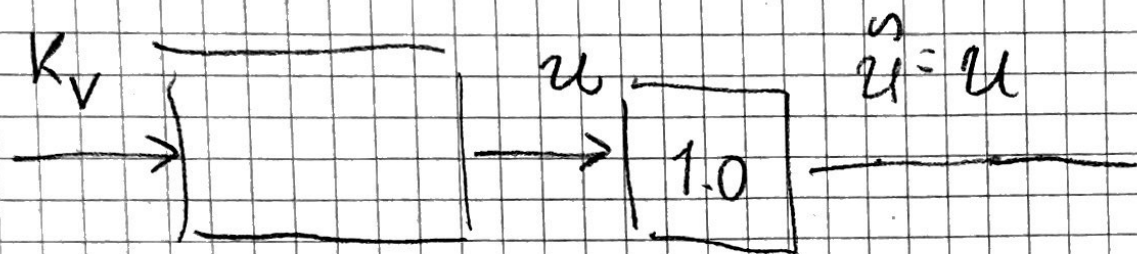
Flow:

$$Q = V_{cyl}^{(ref)} \cdot A, \quad V_{cyl} > 0 \quad \text{⊗} \quad A$$

$$Q = V_{cyl}^{(ref)} \cdot A_a, \quad V_{cyl} < 0 \quad \text{⊗} \quad A_a$$

$$Q < 9.5 \text{ l/min}$$

$U_p = \text{Down}$



$$u = 0,2143 + 0,7857 \cdot \frac{K_v}{3.7796}$$

$U_p:$

$$\text{if } K_v \geq 0.1$$

$$\begin{aligned} U = & -0,013 \cdot K_v^5 + 0,1355 \cdot K_v^4 \\ & - 0,5012 \cdot K_v^3 + 0,7175 \cdot K_v^2 \\ & - 0,0278 \cdot K_v + 0,2684 \end{aligned}$$

$$\text{else if } K_v < 0.1$$

$$U = 0,2325 + 0,0398 \cdot \frac{K_v}{0.1}$$

Check:

$$K_v = 0.1 \Rightarrow U = 0,2723$$

from both sides

Down

2

$$\text{if } K_V \geq 0.1$$

$$\begin{aligned} u = & 0,0054 \cdot K_V^6 - 0,0733 \cdot K_V^5 \\ & + 0,3796 \cdot K_V^4 - 0,9356 \cdot K_V^3 \\ & + 1,0359 \cdot K_V^2 - 0,1046 \cdot K_V \\ & + 0,2381 \end{aligned}$$

$$\text{elseif } K_V < 0.1$$

$$u = 0,2175 + 0,0196 \cdot \frac{K_V}{0.1}$$

Check

$$K_V = 0.1 \Rightarrow u = 0,2371$$

from both sides

sin	100	bar	u pfb	uk
sin	100	bar	m pfb	uk
Ramp	100	bar	u pfb	uk
Ramp	100	bar	m pfb	uk
5	140		u pfb	uk
5	140		m pfb	uk
5	140		u pfb	uk
5	140		m	
5	180		u	
5	180		m	
5	180		u	
5	180		m	

Compensation

Spool

$$\eta = \frac{u + 0.0398}{1.1079}$$

$$u = 0.0095 k_v^3 - 0.1064 \cdot k_v^2 + 0.483 k_v + 0.1528$$

	Q	x	
100	3,71	4,07	
95	3,68	4,06	
90	3,36	3,65	
85	2,81	2,98 3,15	
80	2,34	2,41 2,59	
75	1,86	1,87 1,92	2,16
70	1,56	1,80	
65	1,29	1,50	
60	1,06	1,26	
55	0,91	1,06	
50	0,76	0,91	
45	0,64	0,77	
40	0,53	0,64	
35	0,42	0,51	
30	0,26	0,32	
25	0,06	0,13	

3

spool signal

$$\left. \begin{array}{l} V_{cyl}^{(rel)} > 0 \\ \end{array} \right\} \begin{array}{l} \tilde{u} = \frac{u + 0.0151}{1.1224} \\ 0 \leq \tilde{u} \leq 1 \end{array}$$

$$V_{cyl}^{(rel)} < 0 \quad \text{---}$$

(2)

Valve constant

 $Q > 0$

$$K_V = \frac{Q}{\sqrt{p_s - p_A}}$$

$\frac{l/min}{bar^{0.5}}$

$\swarrow$ Q_{min}
 $\nwarrow$ Q
 $\nearrow$ bar

 $Q < 0$

$$K_V = \frac{|Q|}{\sqrt{p_s - p_B}}$$

make sure $p_s > p_A$ else $K_V = 4$

Real spool pos.

 $V_{cyl}^{(ref)}$ > 0

$$u = -0.013 \cdot K_V^5 + 0.1355 \cdot K_V^4 - 0.5012 \cdot K_V^3 + 0.7175 \cdot K_V^2 - 0.0278 K_V + 0.2684$$

$$0 \leq u \leq 1$$

 $V_{cyl}^{(ref)}$ < 0

Reduser tryk for or knapp !

Sant stant in

